Muhammad Moiz Khalid

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EDUCATION

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES (NUCES - FAST) | Islamabad, Pakistan
BS in Data Science | **Aug 2023 - May 2027** | **CGPA: 3.84/4.0 | Rank 1/150**

- *Gold Medal - Fall '24 & Spring '25* | *Silver Medal - Fall '23 & Spring '26* | *Bronze Medal - Fall '25* | *Dean's List: All Semesters*
- *Relevant Coursework: Agentic AI, Computer Vision, Artificial Intelligence, Intro to Data Science, Database Systems, Advanced Statistics, Data Mining, Data Warehousing & Business Intelligence, Deep Learning, Design & Analysis of Algorithms, Natural Language Processing, Computer Networks, Computer Organization and Assembly Language, Parallel and Distributed Computing*

WORK EXPERIENCE

Researcher Assistant, *Zayed University* | July 2026 - Present

- Collaborate with Dr. Faheem Ullah and researchers from Zayed University and the University of Adelaide on AI and machine learning research, with a focus on big data and cybersecurity.
- Review graduate students' research papers and provide technical feedback to strengthen their work.
- Analyzed user reviews of AI-powered mobile applications to identify permission-related concerns, developing a classifier that achieved 82% accuracy on real-world user-generated data.
- Conducted a systematic literature review on the use of large language models for software vulnerability prediction and prioritization.

Undergraduate Researcher | Computer Vision & Deep Learning, *NUCES - FAST* | Jan 2025 - Present

- Conduct research on efficient deep learning and computer vision, developing models for real-world applications.
- Developed AZNet, a MobileNetV2-based model for Alzheimer's disease classification that achieved 97.8% accuracy with approximately 2.4 million parameters.
- Developed EmoNet, a lightweight speech emotion recognition model combining depthwise separable convolutions, squeeze-and-excitation attention, and a bidirectional LSTM, achieving 96.75% accuracy with 4.11 million parameters.
- Selected as one of five undergraduate researchers based on achieving the highest grade in the supervising professor's course. Published two first-author IEEE conference papers.

Data Annotator, *Turing* | June 2025 - Aug 2025

- Contributed to the development of Claude, Anthropic's AI assistant, by creating and refining high-quality training data for browser-based interactions and reasoning tasks.
- Performed content summarization and fact verification using reliable sources, alongside reasoning-intensive prompt tasks to support model training.
- Evaluated generated data against strict quality standards to ensure factual accuracy and consistency while gaining experience in AI data workflows.

Teaching Assistant / Lab Demonstrator, *NUCES - FAST*

Object Oriented Programming Lab (Fall'24, Spring'26) | **Programming Fundamentals Lab (Spring'25) & Theory (Fall'25)** | **Introduction to Data Science Lab (Fall'25)**

- Mentored 250+ Programming Fundamentals students and assisted with quizzes, assignments, grading, and project supervision.
- Guided 60+ undergraduates in each lab session, providing hands-on support and debugging assistance in C++, OOP, and Python.

PUBLICATIONS

1. **M. M. Khalid**, M. S. Abbas, A. Muhammad, "AZNet: An Efficient Convolutional Neural Network for Alzheimer's Disease Classification," *4th International Conference on Information Science and Communication Technology (ICISCT)*, Karachi, Pakistan, 2026, pp. 1-6, https://doi.org/10.1109/ICISCT67740.2026.11576503.
2. **M. M. Khalid**, A. Rehman, M. O. Yamin, " EmoNet: A Lightweight Attention-Based CNN for Speech Emotion Recognition," *23rd International Bhurban Conference on Applied Sciences & Technology (IBCAST)*, Murree, Pakistan, 2026, pp. 1-6 (Accepted; In Press, https://doi.org/10.48550/arXiv.2610.33977)

3. F. Ullah, B. Shah, M. Watkinson, **M. M. Khalid**, T. K. Khan, M. Junaid "Analysing User Reviews to Identify User Concerns Around Permissions in AI Apps," *6th Conference on Advanced Artificial Intelligence and Education (CAAIE)*, Cardiff, UK, 2026, pp. 1-10 (Accepted; In Press; https://doi.org/10.48550/arXiv.2607.29343)
4. F. Ullah, **M. M. Khalid**, S. A. Khan, H. Ahmad, "LLMs for Software Vulnerability Prediction and Prioritization: A Review " *8th IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD),* Paris, France, 2026, pp. 1-8 (Accepted; In Press)
5. F. Ullah, P. Gupta, **M. M. Khalid**, H. Ahmad, "Security Vulnerabilities in Big Data Systems: An Empirical Study Across 19 Platforms" (Under Review; PeerJ CS Journal)
6. F. Ullah, **M. M. Khalid**, H. Ahmad, "Towards AI-Driven Cyber Deception Mechanism for Enterprise Networks: A Literature Review " (Journal paper; Ready for Submission)
7. F. Ullah, J. Hynds, **M. M. Khalid**, S. A. Khan, "A Machine Learning based Approach for Detecting Failures in Big Data Systems" pp. 1-10 (Conference paper; Ready for Submission)
8. T. Jois, H. Ahmad, F. Ullah, **M. M. Khalid, "**Australian Bushfire Intelligence with AI-Driven Environmental Analytics," (ongoing)
9. **M. M. Khalid**, A. Rehman, M. O. Yamin, "Lightweight Agentic Threat Hunting: A Cost-Aware Escalation Gate for Security Triage," (ongoing)
10. **M. M. Khalid**, A. Rehman, "LiteIDS: A Lightweight Attention-Based Framework for Network Intrusion Detection," (ongoing)

PROJECTS

Case Study: Energy Consumption in London *(Python)*

- Predicted energy consumption behaviors in London based on different demographic, temporal, & weather-based predictors.
- Analyzed 3.5 million observations across different consumers & days and tested 3 classical regression & 4 tree-based models.
- Achieved the lowest MSE of 1.90 with the Random Forest model and identified the ACORN score (demographic type) and max. temperature of the day as the most important factors.

Near-Real-Time Data Warehouse for Walmart using HYBRIDJOIN, *NUCES - FAST*

- Designed a MySQL star-schema warehouse processing 550K transactions (2015–2020), 5,891 customers, and 3,631 products.
- Implemented HYBRIDJOIN using a 10K-slot hash table, doubly-linked queue with O(1) arbitrary-node deletion, and 500-tuple indexed disk partitions.
- Built a multithreaded ETL pipeline with surrogate keys, date attributes, revenue calculations, and transactional batch loading.

Research Document Revision Tracker with Persistent Memory, *NUCES - FAST*

- Built a Groq (gpt-oss)-powered agent to verify reviewer comment resolution across PDF/LaTeX revisions and detect regressions.
- Designed a 23-table SQLite memory system storing document versions, passage-level diffs, evidence, verdicts, and author decisions.
- Added evidence-based verdict validation, prompt-injection safeguards, regression detection, human approvals, transactional writes, and Excel review reports.